

Video Slides W4: Elasticity

ECO 101: Principles of Microeconomics

Prof. Freitas, University of Toronto

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Defining Elasticity

Topic: Elasticity

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Video Learning Objectives

- Why Elasticity?
- Define Elasticity
- Interpret elasticity numbers

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Thought Experiment 1

How large is the effect? Why does it matter?

Should you subsidize preventive health products that have public health benefits (e.g. insecticide-treated bednets, water purification products, and latrine slabs for safe sanitation)?

Abdul Latif Jameel Poverty Action Lab (J-PAL). 2018. "The impact of price on take-up and use of preventive health products." J-PAL Policy Insights. Last modified May 2018.

Thought Experiment 2

How large is the effect? Why does it matter?

When you change the interest rate, how large is the change in the demand for your microcredit loans?

Dean Karlan, Jonathan Zinman, "Long-Run Price Elasticities of Demand for Credit: Evidence from a Countrywide Field Experiment in Mexico", The Review of Economic Studies, Volume 86, Issue 4, July 2019, Pages 1704–1746

Thought Experiment 3

How large is the effect? Why does it matter?

An animal welfare proposition increases the cost of producing pork. How large is the change in quantity of pork sold in California?

UC Davis Professor Says Impact of Proposition 12 Won't Be Catastrophic. By Jennifer Shike on Farm Journal, agweb.com. September 14, 2021

How Large is the Effect?

When price changes how “big” is the quantity response?

- Problem: units matter
 - is \$1 a big change? \$1=61 Rupees
 - 1 gallon=3785.41 ml
- Solution: Think about percentage changes.

Elasticity

Notation: Δ is change, $\% \Delta$ is percentage change

- Elasticity is unit free measure of how responsive quantity is to a change in something.
- Elasticity in general is

$$\frac{\% \Delta Q}{\% \Delta \text{something else}}$$

- Sign tells us if the numerator and denominator work in the same or opposite directions.

(Own) Price Elasticity of Demand

Δ : change, $\% \Delta$: percentage change,

- Elasticity: compare the percentage change in quantity demanded (i.e. $\% \Delta Q^D$) to the percentage change in price (i.e. $\% \Delta P$).

$$\varepsilon_D = \frac{\% \Delta Q^D}{\% \Delta P}$$

Absolute Elasticity of Demand

Δ : change, $\%\Delta$: percentage change, $\varepsilon_D = \frac{\%\Delta Q^D}{\%\Delta P}$

- Usually price and quantity demanded move in opposite directions.
 - negative sign not informative
- For simplicity, typically use absolute value when discussing elasticity of demand.
- General rule: Keep sign on elasticity if informative.

Interpreting an Elasticity Number

Δ : change, $\%\Delta$: percentage change, $|\varepsilon_D| = \left| \frac{\%\Delta Q^D}{\%\Delta P} \right| = 0.8 = \frac{4}{5}$

- Given a $1\%\Delta$ in P what is the $\%\Delta$ in Q^D ?
 $1\%\Delta \uparrow P \Rightarrow Q^D$

$$|\varepsilon_D| = \left| \frac{\%\Delta Q^D}{\%\Delta P} \right|$$

- If market Q^D changes by $2\%\Delta$, by how much does P have to change?
 - $2\%\Delta \downarrow Q^D \Rightarrow P$

Summary

- Elasticity is a unit free measure of quantity responsiveness.
- General formula

$$\frac{\%\Delta Q}{\%\Delta \text{something else}}$$

- We use the absolute value for demand elasticity only.

Calculating Elasticity

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Video Learning Objectives

Calculate elasticity numbers using the

- Mid-point method from two points and
- The point method from a linear demand/supply function.

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Percentage Change

Notation: Δ is change, $\% \Delta$ is percentage change

- Change in quantity is

$$\Delta Q = Q_2 - Q_1$$

- Percentage change in quantity is

$$\% \Delta Q =$$

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Calculating Elasticity

Example: ε_D : (own-price) Elasticity of demand

Two ways depending on the information you have

- Discrete data $\{Q_A^D, P_A\}$ and $\{Q_B^D, P_B\}$
- Continuous data e.g. $P(Q^D) = 200 - 2Q$

Midpoint method: Calculating $\% \Delta$

$\% \Delta$: **percentage change**

Example: Given $\{Q_A^D = 8, P_A\}$ and $\{Q_B^D = 10, P_B\}$

- $8 \rightarrow 10 \Rightarrow |\% \Delta| = \left| \frac{8-10}{8} \right| = \frac{2}{8}$
- $10 \rightarrow 8 \Rightarrow |\% \Delta| = \left| \frac{10-8}{10} \right| = \frac{2}{10}$

Solution: mid point method

- $\left| \frac{10-8}{\text{avg } Q} \right| = \frac{2}{9}$

Midpoint method: Calculating ε_D

$\{Q_A^D = 8, P_A = \$3.50\}$ and $\{Q_B^D = 10, P_B = \$2.50\}$

$$\left| \frac{Q_A^D - Q_B^D}{\text{avg } Q} \right| =$$

$$\left| \frac{P_A - P_B}{\text{avg } P} \right| =$$

$$|\varepsilon_D| =$$

Point elasticity: Calculating ε_D

$$Q^D = 100 - 0.5P \text{ or } P = 200 - 2Q^D$$

- $|\varepsilon_D| = \left| \frac{\% \Delta Q^D}{\% \Delta P} \right|$
- Note: $\% \Delta Q^D = \frac{\Delta Q^D}{Q^D}$, $\% \Delta P = \frac{\Delta P}{P}$

Point elasticity: Calculating ε_D

$$Q^D = 100 - 0.5P \text{ or } P = 200 - 2Q^D$$

- $|\varepsilon_D| = \left| \frac{\frac{\Delta Q^D}{Q^D}}{\frac{\Delta P}{P}} \right| = \left| \frac{\Delta Q^D}{\Delta P} \frac{P}{Q^D} \right|$
- We know $Q^D = 100 - 0.5P$ so $\Delta Q^D =$

What method to use?

- Discrete data: Mid-point method.

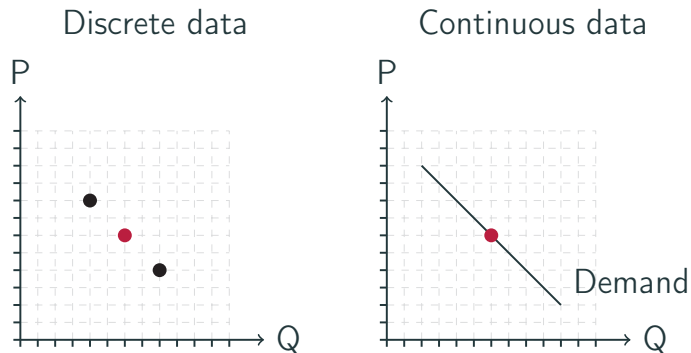
$$|\varepsilon_D| = \left| \frac{\frac{\Delta Q^D}{\text{avg } Q^D}}{\frac{\Delta P}{\text{avg } P}} \right|$$

- Continuous data: Point method.

$$|\varepsilon_D| = \left| \frac{\Delta Q^D}{\Delta P} \frac{P}{Q^D} \right|$$

What method to use?

Goal: calculate elasticity at a given point.



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Summary

- Data availability determines which method to use when calculating elasticity.
- Mid-point method when you have discrete data and the Point method when you have continuous data in the form of an equation.

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Elasticity: Types and Classification

Topic: Elasticity

Video Learning Objectives

- Define and explain elasticity categories
 - unit elastic, elastic and inelastic
 - perfectly elastic and perfectly inelastic
- Define different types of elasticity and use them to classify goods

(Own) Price Elasticity of Demand

$\% \Delta$: **percentage change**, $|\varepsilon_D| = \left| \frac{\% \Delta Q^D}{\% \Delta P} \right|$

Elasticity: compare the percentage change in quantity demanded (i.e. $\% \Delta Q^D$) to the percentage change in price (i.e. $\% \Delta P$).

(Own) Price Elasticity of Demand

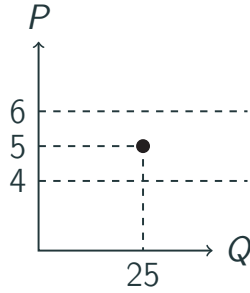
$\% \Delta$: **percentage change**, $|\varepsilon_D| = \left| \frac{\% \Delta Q^D}{\% \Delta P} \right|$

- Elastic demand: $|\% \Delta Q^D| > |\% \Delta P| \Rightarrow |\varepsilon_D| > 1$
- Inelastic demand: $|\% \Delta Q^D| < |\% \Delta P| \Rightarrow |\varepsilon_D| < 1$
- Unit elastic: $|\% \Delta Q^D| = |\% \Delta P| \Rightarrow |\varepsilon_D| = 1$

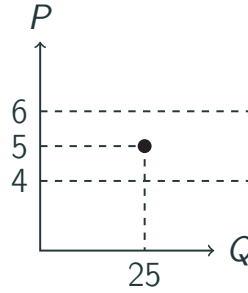
(Own) Price Elasticity of Demand

$\% \Delta$: **percentage change**, $|\varepsilon_D| = \left| \frac{\% \Delta Q^D}{\% \Delta P} \right|$

Elastic Demand



Inelastic Demand



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Extreme Elasticities

$\% \Delta$: **percentage change**, $|\varepsilon| = \left| \frac{\% \Delta Q}{\% \Delta \text{something else}} \right|$

Perfectly Elastic

$$|\varepsilon| = \infty$$



Perfectly Inelastic

$$|\varepsilon| = 0$$



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More Elasticities

Notation: $\% \Delta$ is percentage change, $\varepsilon = \frac{\% \Delta Q}{\% \Delta \text{something else}}$

Something else...

- (Own) price
- Price of other goods
- Income
- Supply

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Supply Elasticity

Supply Elasticity

$$\varepsilon_S = \frac{\% \Delta Q_S}{\% \Delta P}$$

Cross-Price Elasticity

- How quantity demanded for good A changes when the price of good B changes

$$\varepsilon_{CP} = \frac{\% \Delta Q_A}{\% \Delta P_B}$$

- A & B are substitutes: ε_{CP}
- A & B are complements : ε_{CP}

Income Elasticity

- Income Elasticity $\varepsilon_I = \frac{\% \Delta Q}{\% \Delta I}$
- Normal goods $\varepsilon_I > 0$
- Inferior goods $\varepsilon_I < 0$

Summary

- Elastic demand is when the % change in quantity demanded is larger than the % change in price.
- Different types of elasticity capture how responsive consumers are to changes in their environment.
- Use the elasticity sign to help classify goods.

Elasticity and Expenditures

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Video Learning Objectives

- Explain how expenditures changes with price elasticity of demand
- Explain how and why elasticity changes along a linear demand curve.

Expenditure and ε_D

$$\text{Expenditure} = \text{Revenue} = P * Q \neq \text{Profit} = \text{Revenue} - \text{Costs}$$

Price increases changes a seller's revenues

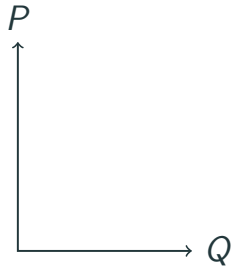
1. For the units sold, higher price per unit.
2. But, fewer units sold.

Elasticity	P	$*$	Q	= Total Revenue
Elastic		*		
Unit elastic		*		
Inelastic		*		

Expenditure and ε_D

$$\text{Expenditure} = \text{Revenue} = P * Q \neq \text{Profit} = \text{Revenue} - \text{Costs}$$

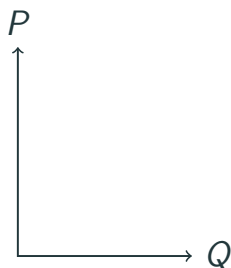
If Netflix raised its prices, would its revenues increase or decrease?



Expenditure and ε_D

$$\text{Expenditure} = \text{Revenue} = P * Q \neq \text{Profit} = \text{Revenue} - \text{Costs}$$

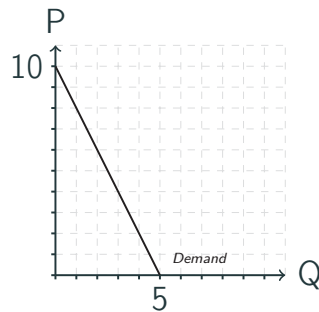
If the price of cocoa increases, would consumer expenditures on chocolate increase or decrease?



Elasticity along a linear demand curve

$\% \Delta$ is percentage change, $|\varepsilon_D| = \left| \frac{\% \Delta Q^D}{\% \Delta P} \right| = \left| \frac{\Delta Q^D}{\Delta P} \cdot \frac{P}{Q^D} \right|$

$$Q^D = 5 - 0.5P$$



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Summary

- Elasticity can predict how total revenue changes with a change in price.
- Elasticity changes along a linear demand curve.

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Determinants of Elasticity

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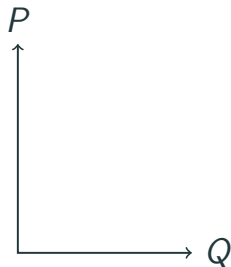
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Video Learning Objectives

- Identify some factors that could explain why the quantity response to a change is elastic or inelastic

Elasticity of Demand

- When the price rises, quantity demanded decreases.
- For a 10% increase in prices, what determines how large the quantity response is?



Elasticity of Demand

- To understand a consumer's quantity response to a price change we typically look at the consumer's willingness and ability to substitute away from (to) the good at that price.

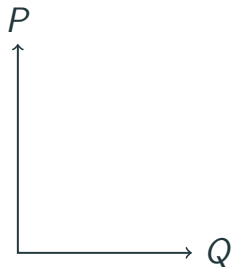
Elasticity of Demand

ε_D : (own-price) Elasticity of demand

- Luxuries vs. necessities
- Long-run vs. short-run
- Narrow market vs. broad market

Elasticity of Supply

- When the price rises, quantity supplied increases.
- For a 10% increase in prices, what determines how large the quantity response is?



Elasticity of Supply

- When price rises, the quantity response depends on the firms ability to increase output at that price.

Elasticity of Supply

Q: How easily can suppliers get inputs?

Easier to find inputs $\Rightarrow$ larger quantity response to a given price change

- Input specificity
- Input availability
- Time frame

Summary

- How responsive consumers or producers are when prices rise depends on what they can and are willing to do in response to the higher price.